Cryptographic Challenges and Security in Post Quantum Cryptography Migration: A Prospective Approach

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***Abstract*—Post-Quantum Cryptography (PQC) is a rapidly developing field that addresses the potential threat quantum computers pose to the security of current cryptographic algorithms. With the rapid advancement of quantum computing technologies, it has become imperative to identify and standardize new cryptographic techniques that can withstand attacks from quantum computers. This paper explores the motivation behind PQC and provides an overview of the experimental research conducted in the field. It highlights the vulnerabilities of traditional cryptographic algorithms, such as RSA and ECC, to quantum attacks and emphasizes the need for transitioning to quantum-resistant alternatives. The experiments conducted in the PQC domain encompass the evaluation and analysis of candidate algorithms submitted to the NIST PQC competition. These experiments involve comprehensive security assessments, performance evaluations, and code audits to ensure the suitability and effectiveness of the proposed algorithms in real-world scenarios. This paper provides guidelines for organizations and practitioners adopting PQC. It discusses considerations for selecting and implementing post-quantum cryptographic algorithms, including security levels, algorithmic characteristics, performance trade-offs, and integration challenges. It also outlines the importance of a smooth transition from classical to post-quantum cryptographic systems and highlights the need for standardization efforts to ensure interoperability and compatibility.**

***Index Terms*—Public Key Infrastructure, Post Quantum Cryptography, Shor Algorithm, Lattice-Based Cryptography, Hash-based Algorithms**

I. Introduction

Security experts worldwide are warning that with the rapid development in Quantum Computers and Quantum Algorithms, in a few years, current cryptography techniques based on public key encryption like RSA, ECC and public key exchange of DH and ECDH and their variant public-key exchange algorithms shall be broken, and the strength of symmetric algorithms like AES shall reduce to half. This implies that all applications that use IPSec, TLS and SSL for Secure Communications are at risk. Post-Quantum-Cryptography (PQC) refers to new algorithms that shall be safe from the threat of Quantum Computers [1]. NIST in the USA is doing standardization of PQC algorithms. Draft PQC standards shall come out sometime in 2024. Currently, in round 3 of the NIST PQC Competition, Crystals-Dilithium, FALCON, and SPHINCS+ have moved forward in digital signature algorithms, while Crystals-Kyber has been selected for Public-key Encryption and Key-establishment algorithm [2].

A. Threats to Asymmetric Cryptography

Shor's algorithm [3] poses a significant threat to asymmetric cryptography, especially to cryptosystems that rely on the hardness of factoring large integers and the discrete logarithm problem. Asymmetric cryptography, also known as public-key cryptography, is widely used for secure communication, digital signatures, and other cryptographic operations. The two most common cryptographic algorithms affected by Shor's algorithm are RSA (Rivest-Shamir-Adleman) and certain variants of the Diffie-Hellman key exchange like Elliptic Curve Diffie-Hellman.

Shor's quantum algorithm can efficiently factor large integers and compute discrete logarithms on a quantum computer in polynomial time. It uses the principles of quantum computing, such as superposition and entanglement, to perform these computations exponentially faster than classical algorithms. A quantum computer could break the security of many commonly used asymmetric cryptographic schemes if powerful enough.

B. Threats to Symmetric Cryptography

Grover's algorithm [4] poses a threat to symmetric cryptography, as it is a quantum algorithm that can efficiently search through an unsorted database with N entries in approximately $\sqrt{N}$ time. This means Grover's algorithm can significantly speed up brute-forcing symmetric encryption keys, typically much shorter than those used in asymmetric cryptography.

Symmetric cryptography uses a shared secret key to encrypt and decrypt data. The security of symmetric encryption algorithms is based on the assumption that an attacker must try all possible keys (brute force) to recover the original data from the ciphertext. The number of possible keys is directly related to the length of the key.

With classical computers, the time complexity to search through all possible keys (brute-force attack) in symmetric

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cryptography is proportional to 2^N, where N is the length of the key in bits. This exponential growth in possible keys makes brute forcing impractical, especially for long keys.

However, Grover's algorithm can search through the space of possible keys with a time complexity of approximately $\sqrt{N}$. This reduces the computational effort required to brute-force the key from exponential to quadratic, which is a significant improvement. (see Table I) For example, if the symmetric encryption key has a length of 128 bits, Grover's algorithm can find the key in roughly 2^{64} iterations, making the attack much faster than a classical brute-force attack.

TABLE I
EFFECTIVE KEY SIZE OF CRYPTOGRAPHY SCHEMES

Algorithm	Key Size	Effective Key Size in Classical Computing	Effective Key Size in Quantum Computing
RSA-1024	1024	80	0
RSA-2048	2048	112	0
ECC-256	256	128	0
ECC-384	384	256	0
AES-128	128	128	64
AES-256	256	256	128

C. Threats to Hash Algorithms

Grover's algorithm also threatens hash algorithms, as it can speed up finding collisions in hash functions. A collision occurs when two different inputs produce the same hash value, and the security of hash functions is built on the assumption that finding collisions is computationally infeasible. Algorithms like SHA1 and SHA2 are widely used across many critical applications, so a Quantum-Safe hash is needed to replace it. (See Table II)

With classical computers, finding collisions in hash functions generally requires an exhaustive search, which has a time complexity of approximately $2^{n/2}$, where 'n' is the length of the hash output in bits. This exponential growth in possible hash values makes it computationally infeasible to find collisions for a robust cryptographic hash function with sufficiently long output lengths.

However, Grover's algorithm can search for collisions in a quantum superposition state and reduce the time complexity to approximately $2^{n/3}$, a quadratic speedup compared to classical methods. For example, for a hash output length of 256 bits, Grover's algorithm can find collisions in approximately $2^{256/3}$ iterations, making the search more efficient than classical brute-force methods.

II. THE OVERVIEW OF POST-QUANTUM CRYPTOGRAPHY

Post-Quantum Cryptography (PQC) is a field of research focused on developing cryptographic algorithms that can resist attacks from powerful quantum computers. The following are the most prominent and promising approaches within the field of PQC:

TABLE II
SECURITY OF CRYPTOGRAPHY SCHEMES IN QUANTUM WORLD

Algorithm	Status in Post Quantum World
AES-256	Secure
SHA-256, SHA-3	Secure
RSA	Not Secure
ECDSA, ECDH	Not Secure
DSA	Not Secure

A. Lattice-based Cryptography

Lattice-based cryptography [5] is a type of cryptographic scheme that relies on the mathematical properties of lattices, which are geometric structures in multi-dimensional spaces. The security of lattice-based cryptography is based on the hardness of some computational problems associated with lattices.

In a lattice, points are arranged in a periodic pattern, forming a grid-like structure. The fundamental problem that underpins lattice-based cryptography is the Shortest Vector Problem (SVP) and its related variants, such as the Learning With Errors (LWE) problem and the Ring Learning With Errors (RLWE) problem.

The LWE problem involves finding a short lattice vector close to a random linear combination of other lattice vectors with some noise. The presence of noise makes it challenging to extract useful information and solve the problem efficiently. This inherent difficulty forms the basis for cryptographic security.

B. Code-based Cryptography

Code-based cryptography [6] is a form of public-key cryptography that relies on the hardness of certain decoding problems related to error-correcting codes. The security of code-based cryptographic schemes is based on the assumption that certain decoding problems are computationally infeasible to solve efficiently, even for powerful adversaries. A random binary irreducible Goppa code serves as the private key, while a random generator matrix of a randomly permuted version of that code makes up the public key. The cipher-text being utilized here is a codeword with added errors.

The fundamental building block of code-based cryptography is the error-correcting code, which encodes messages so that errors introduced during transmission can be corrected at the receiver's end. The key idea in code-based cryptography is to exploit the difficulty of decoding a received message to recover the original encoded message without knowing the private key.

C. Multivariate Cryptography

Multivariate cryptography [7] is a form of public-key cryptography based on the hardness of solving systems of multivariate polynomial equations over finite fields. The security of multivariate cryptographic schemes relies on the difficulty of solving these polynomial equations, known to be NP-hard (computationally difficult). The public and private keys in

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multivariate cryptography consist of multivariate polynomials and their inverses.

D. Hash-based Cryptography

Hash-based cryptography [8], [9]is a post-quantum cryptographic scheme that relies on the properties of cryptographic hash functions. Unlike traditional public-key cryptosystems based on mathematical problems like factoring and discrete logarithms, hash-based cryptography derives its security from the collision resistance and one-wayness properties of hash functions.

Hash functions take an arbitrary length input and produce a fixed-size output, often called a hash value or digest. Hash-based cryptographic schemes use hash functions as the building block for various cryptographic primitives, such as digital signatures and public-key encryption. These schemes often involve time-memory trade-offs and rely on the assumption that finding collisions in hash functions is difficult.

One of the well-known hash-based cryptographic schemes is the Merkle signature scheme, which provides a digital signature scheme based on hash functions and time-memory trade-offs. Another example is the Lamport signature scheme, which uses one-way hash chains to generate a digital signature.

III. NIST PQC Competition

The NIST PQC (Post-Quantum Cryptography) competition is an initiative launched by the National Institute of Standards and Technology (NIST) in 2016 to standardize new cryptographic algorithms that can resist attacks from quantum computers. Quantum computers have the potential to break many of the currently used cryptographic algorithms, such as RSA and ECC, which form the basis of secure communications on the internet.

The NIST PQC competition aims to identify and select quantum-resistant cryptographic algorithms that can replace existing ones and ensure long-term security in a post-quantum era. The competition consists of several rounds involving the cryptographic community's evaluation and analysis of submitted candidate algorithms.

The competition began with an initial call for proposals, which concluded in November 2017. NIST received a significant number of submissions, and these algorithms underwent extensive scrutiny and evaluation by experts in the field. NIST then selected several algorithms as finalists, announced in January 2019.

The NIST PQC competition aims to foster collaboration and transparency in the cryptographic community, as experts worldwide contribute to the evaluation process. Currently, in round 3 of the NIST PQC Competition, Crystals-Dilithium, FALCON, and SPHINCS+ have moved forward in digital signature algorithms, while Crystals-Kyber has been selected for Public-key Encryption and Key-establishment algorithm [2]. The selection of the final set of standardized post-quantum algorithms by NIST provides a foundation for organizations to transition to quantum-resistant cryptography as quantum computers become more prevalent.

IV. PQC Selected Algorithms

A. Public-key Encryption and Key-establishment Algorithms

1) Crystals-Kyber [10]: It is a lattice-based cryptographic algorithm designed to resist attacks from classical and quantum computers. The CRYSTALS-Kyber submission was made by researchers from several institutions, including the University of Tartu, Inria, and NXP Semiconductors.

Kyber is an IND-CCA2-secure key encapsulation mechanism (KEM), whose security is based on the hardness of solving the learning-with-errors (LWE) problem over module lattices. Kyber is one of the candidate algorithms submitted to the NIST post-quantum cryptography project. The submission lists three different parameter sets aimed at different security levels. Specifically, Kyber-512 aims at security roughly equivalent to AES-128, Kyber-768 aims at security roughly equivalent to AES-192, and Kyber-1024 aims at security roughly equivalent to AES-256.

Strength

For the majority of applications, the scheme performs superbly all around. Adjusting the module rank and noise parameters enables performance/security trade-offs.

Weakness

1) KYBER doesn't use the public-key compression present in the first round version for round 2.
2) A new fault attack forcing nonce reuse was discovered during the second round, affecting several lattice schemes, including KYBER.

B. Digital Signature Algorithms

1) CRYSTALS-DILITHIUM: The "Cryptographic Suite for Algebraic Lattices" (CRYSTALS) encompasses two cryptographic primitives: Kyber, an IND-CCA2-secure key-encapsulation mechanism (KEM); and Dilithium, a strongly EUF-CMA-secure digital signature algorithm. Both algorithms are based on hard problems over module lattices, are designed to withstand attacks by large quantum computers, and have been submitted to the NIST post-quantum cryptography project.

Strength

1) For the second round, DILITHIUM added the option to generate a signature non-deterministically and added an implementation based on using AES rather than SHAKE to illustrate the future benefits of hardware instructions
2) Overall, DILITHIUM has strong, balanced performance in terms of key and signature sizes and in the efficiency of the key generation, signing, and verification algorithms.
3) DILITHIUM performs well in real-world experiments. The applications which are fully fit are:
   a) CA Keys
   b) TLS Client
   c) TLS Server
   d) Bitcoin
   e) FIDO
   f) PGP

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g) Code Sign
h) PDF-AES

Weakness

DILITHIUM has the lowest CoreSVP security strength parameter set of any of the lattice schemes still in the process The applications which are not fit are:

1) EMV-DDA
2) GSM eSIM

2) FALCON: Falcon (Fast Fourier Lattice-Based Compact Signatures over NRTU) [11], [12] was submitted on 30th November 2017 to the NIST PQC Project. FALCON is based on the hardness of the Shortest Vector Problem (SVP) in lattice theory. In lattice-based cryptography, the scheme's security relies on the difficulty of finding the shortest non-zero vector in a lattice, a geometric structure in multi-dimensional space. It is based on the theoretical framework of Gentry, Peikert and Vaikuntanathan for lattice-based signature schemes.

Strength

1) The smallest bandwidth (public key size and signature size) of all of the second-round digital signature schemes, although key generation is slower
2) FALCON can easily be put into existing protocols and applications and offers very good overall performance. The applications which are fully fit are:
   a) CA Keys
   b) TLS Client
   c) TLS Server
   d) Bitcoin
   e) FIDO
   f) PGP
   g) Code Sign
   h) PDF-AES

Weakness

FALCON's category one parameters have relatively low CoreSVP security strength. The partially fit applications are:

1) 3SKey
2) EMV-SDA
3) EMV-DDA
4) ICAO9303
5) GSM eSIM
6) PDF-QE

3) SPHINCS+: SPHINCS+ (SPHINCS Plus) [13], submitted to the NIST PQC project, is a state-of-the-art post-quantum cryptographic scheme based on the concept of hash-based signatures. It is designed to provide secure digital signatures resistant to classical and quantum attacks. SPHINCS+ is an evolution of the original SPHINCS signature scheme, presented at EUROCRYPT 2015. It aims at reducing the signature size by making several improvements. [9], [13] may be referred to for an overview of the major improvements from SPHINCS to SPHINCS+.

Strength

1) A stateless hash-based signature scheme
2) A very small key size of – 32 to 64 bytes, depending on the security level

Weakness

1) Signing is very slow, and signatures are relatively large, although SPHINCS+ offers different parameter choices that trade-off signature size for speed.
2) The design of SPHINCS+ makes it especially vulnerable to fault attacks and, to a lesser extent, side-channel attacks.
3) Will not work if put into existing protocol and applications.

Further, four KEM candidates which need more evaluation are advanced to Round 4 of the competition. The candidates who are advanced to Round 4 are as follows: (a) Classic McEliece (b) BIKE (Bit Flipping Key Encapsulation), (c) HQC (Hamming Quasi Cycle), and (d) SIKE (supersingular isogeny key encapsulation).

V. Limitations of PQC

PQC has certain limitations and challenges that need to be addressed. Some of the key limitations of PQC include:

A. Performance Overhead

Many PQC algorithms have higher computational requirements compared to traditional cryptographic algorithms. This increased performance overhead can impact the efficiency and responsiveness of cryptographic operations, especially on resource-constrained devices.

B. Key Sizes

Some PQC algorithms require larger key sizes than their classical counterparts to achieve the same level of security. Larger key sizes can lead to increased bandwidth and storage requirements, affecting the overall performance and usability of PQC.

C. Lack of Standardization

While there are promising PQC candidates, the standardization process for selecting and certifying post-quantum cryptographic algorithms is still ongoing. The lack of standardized algorithms may introduce interoperability issues and uncertainties in deploying PQC in real-world systems.

D. Implementation Complexity

Implementing PQC algorithms correctly can be challenging. Insecure or incorrect implementations could lead to vulnerabilities that compromise the security of the entire cryptographic system. Ensuring proper implementation and adherence to best practices is crucial for PQC adoption.

E. Quantum Computing Progress

The rate of progress in quantum computing technology and the development of more powerful quantum computers could potentially render specific PQC algorithms insecure before they become widely adopted. The dynamic nature of quantum computing progress requires constant vigilance and updates in choosing PQC algorithms.

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F. Cryptanalysis

As with any cryptographic system, PQC algorithms are subject to cryptanalytic attacks. While PQC candidates undergo extensive analysis, new vulnerabilities or attack techniques might be discovered in the future, necessitating ongoing research and analysis of PQC algorithms.

G. Transition Complexity

Migrating from classical cryptographic systems to PQC can be complex, especially in large-scale systems with legacy infrastructure and dependencies on existing cryptographic protocols. Organizations must plan a smooth transition to avoid disruptions and ensure secure operations.

H. Limited Deployment Experience

PQC is still relatively new, with limited real-world deployment experience compared to classical cryptographic systems. This lack of historical data may introduce uncertainties about the long-term security and performance of PQC.

VI. International Efforts in Finalizing PQC Standards

European and other Western countries have aligned themselves with NIST for PQC standardization but are planning to have their own implementation of NIST-standardized PQC algorithms/protocols. China and South Korea have their own PQC standardization bodies. China has already completed the PQC algorithm/protocol selection process and has shortlisted the winners. On the other hand, Japan has aligned with the NIST process and has been rigorously participating in the competition. Apart from NIST (US), the following organizations and nations are involved in the Post Quantum Cryptographic standardizations/adoptions:

A. ETSI (European Telecommunication Standards Institute)

ETSI published two technical reports in October 2021 about "Quantum-Safe Signatures" and "Quantum-Safe Public Key Encryption and Key Encapsulation", which support NIST PQC standards [14], [15].

B. ISO (International Organization for Standardization)

ISO/IEC JTC 1/SC 27/WG 2 and ISO/TC 68/SC 2/WG 11 are working to finalize Post Quantum Cryptography for long-term security but have not published or shortlisted anything in the public domain as of date.

C. IETF (Internet Engineering Task Force)

IETF focuses on developing Internet protocol standards. Discussions are being held to establish a working group (IETF 115, November 2022) to provide transition support to PQC in protocols. Recently there have been proposals for specifying algorithm identifiers and ASN.1 encoding format for Kyber (Key encapsulation mechanism) in Public Key certificates [7]. Similar efforts are also in place for Cryptographic Message Syntax (CMS). The usage of Dilithium (Digital Signature Scheme) in X.509 certificates and CRLs are also currently in discussion [8].

D. Post Quantum Efforts in Japan

CRYPTREC is the Cryptography Research and Evaluation Committee set up by the Japanese government to evaluate and recommend cryptographic techniques for government and industrial usage [16]. In 2013, the CRYPTREC cipher list was released, and in 2014 the CRYPTREC Report on the difficulty of lattice problems was released. In addition, the Cryptanalysis Evaluation Working Group got established in 2015. In 2018, this organization released a report on PQC. Japan now seems to have aligned with NIST PQC standardization efforts, and their researchers have contributed to the NIST standardization process. Japan has also come up with a PQC-enabled smart card. The following are some of Japan's efforts in adopting Post Quantum Cryptography. In October 2022, PQC CARD, the world's first smart card with post-quantum cryptography, was created by Toppan and the National Institute of Information and Communications Technology (NICT) Japan [17]. They demonstrated the effectiveness of applying PQC CARD to manage access to H-LINCOS, a system for the safe long-term storage and transmission of healthcare data. CRYSTALS-Dilithium has been used as a Digital Signature Scheme in the PQC CARD. The NIST round one PQC standardization competition had many contributions from Japanese Researchers. Some of the contributors include the following:

- Classic McEliece: includes. Chou (Osaka U.)
- Ding Key Exchange: includes T. Takagi, Y. Wang (UT, Kyushu U.)
- Giophantus: K. Akiyama, H. Shimizu (Toshiba) , Y. Goto (HUE), S.
- Okumura (Osaka U.), T. Takagi, Y. Ikematsu (Kyushu U.), K. Nuida
- Goichiro Hanaoka - Advanced Cryptography Research Team
- LOTUS: L. T. Phong, T. Hayashi, Y. Aono, S. Moriai (NICT)
- RaCoSS: incl. K. Fukushima, P.S. Roy, R. Xu, S. Kiyomoto (KDDI), T.
- Takagi (UT)

The " McEliece PQC" algorithm was also the R3 finalist.

E. PQC Implementation in the United Kingdom

The National Cyber Security Centre (NCSC) of the UK anticipates that once NIST standards are ready and protocols (IPSec, TLS, etc.) are modified to accommodate Quantum-Safe Cryptography (QSC), significant commercial goods and services will migrate to QSC [18]. Most users' recommended course of action is to adhere to industry-standard cyber security best practices and defer developing QSC solutions that comply with standards. Additionally, NCSC advises organizations in charge of their own cryptographic infrastructure to make the quantum-safe transition a part of their long-term plans and to research which of their systems will require transition at the top of the priority list. Priority systems may include those that handle delicate personal information or those components of the public-key infrastructure with

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certificate expiration dates that are far in the future and would be challenging to replace.

The NCSC also warns against implementing non-standard QSC too soon. If the product is not based on a standardized algorithm, the security level might not be verifiable, and such items are unlikely to function with future standards-compliant products. This can result in a second unneeded transition to a QSC that complies with standards.

F. China PQC Efforts

The Chinese Association for Cryptologic Research (CACR) started the Post Quantum Cryptography standardization process in 2018 and shortlisted three candidates in January 2020. Aigi-sig, LAC.PKE and Aigis-enc are the shortlisted candidates, of which Aigi-sig is a digital signature scheme, while LAC.PKE and Aigis-enc are KEMs; All of them are based on lattice schemes. [19], [20]

G. Republic of Korea PQC Efforts: KpqC contest

South Korea has been standardizing Post Quantum Cryptographic Algorithms through their "National Contest for Quantum Resistant Cryptography" ("KpqC contest") [21]. The request for proposal was announced on Nov 2021. The first round of competition is in progress from November 2022 to November 2023. The contest has received seven Public Key Encryption and Key Establishment algorithms and nine Digital Signature Algorithms. The first-round winners are expected to be announced in November 2023.

H. Notable Efforts by Microsoft, Google, Infineon, and IBM

Microsoft has been working on the development of new schemes and software libraries for post-quantum cryptography. Microsoft is developing four potential cryptography solutions that can survive the power of quantum computers and yet function with current protocols. It is participating in "liboqs" [22] and Open Quantum Safe project, which is a software prototyping platform for quantum-resistant cryptography [23]. Further, Microsoft is working on Post-Quantum Crypto VPN, which is a fork of the OpenVPN project. In 2016, Google experimentally deployed the post-quantum key-exchange scheme "New Hope" for connections between the Chrome browser (client machines) and Google servers [24]. In 2017, Infineon succeeded in implementing a variant of "New Hope" on a contactless smartcard microcontroller. The New Hope key exchange protects the smart card and reader communication. This is one of the first commercially available chipsets for PQC for embedded systems [25]. IBM focuses on lattice-based cryptographic solutions, and its research objectives focus on creating useable cryptographic solutions that are immune to the dangers that quantum computing poses. [26].

VII. Performance Evaluation

Table III contains the evaluation result of various PQC algorithms in terms of key pair generation, signing and verifying.

TABLE III
PERFORMANCE OF SIGNING ALGORITHMS FOR DIFFERENT ACTIONS (IN MICRO SECONDS)

Algorithm	Key Pair	Sign	Verify
Dilithium5	54.04	109.58	51.88
Dilithium5-AES	25.75	67.470	26.03
Falcon-1024	14614.9	428.06	71.31
Rainbow-V-Classic	3887097.0	26900.52	27494.88
Rainbow-V-Compressed	4387051.0	2058237.0	29299.26
Picnic_L5_Full	3.28	2665.46	2161.48
Picnic_L5	3.243	14891.45	10952.642
SPHINCS+-SHAKE256-256s-simple	49050.71	598906.16	1591.77
SPHINCS+-Haraka-192f-robust	267.39	7863.68	522.26
SPHINCS+-SHA256-192f-robust	1066.51	31221.32	4472.67

VIII. PQC tutorial using Open Quantum Safe (OQS) Provider

This section describes a set of commands to generate key pairs and certificates using CRYSTAL-Dilithium. The environment uses Open Quantum Safe (OQS) provider [22] and OpenSSL to run the commands. For the digital signature process, variations of Dilithium can be used, details of which are available in table IV.

A. Installation of OQS

A detailed installation guide, along with sample files and screenshots, is available at [27].

B. Generate Key Pairs

- Generate Private Key using CRYSTALS-Dilithium5

```
$ openssl genpkey -algorithm dilithium5 -
    out private.key
```

- Generate Public Key

```
$ openssl pkey -in private.key -pubout -
    out public.key
```

C. Generating Self Signed Certificate

- Generate Self-Signed Certificate

```
$ openssl req -key private.key -new -x509
    -days 365 -out certv3.crt
```

D. Generating (Certifying Authority (CA) Signed Certificate

- Generate CA Private Key (RSA2048) and Certificate

```
$ openssl genrsa -out ca-pvt.key 2048
$ openssl req -new -x509 -key ca-pvt.key -
    days 3650 -out ca.crt
```

- Generate CSR for end user

```
$ openssl req -key private.key -new -out
    certreq.csr -extensions v3_req
```

- Generate Certificate using CSR

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```
$ openssl x509 -req -in certreq.csr -CA ca
    .crt -CAkey ca-pvt.key -CAcreateserial
    -out causer.crt -days 365 -sha256 -
    extensions v3_req -extfile /usr/lib/ssl
    /openssl.cnf
```

A graphical tutorial for all the above operations is available at [27].

TABLE IV
DILITHIUM PARAMETERS SUMMARY

Algorithm	NIST level	Public Key Size (bytes)	Private Key Size (bytes)	Signature Size(bytes)
Dilithium2	2	1312	2528	2420
Dilithium3	3	1952	4000	3293
Dilithium5	5	2592	4864	4595

IX. GUIDELINES AND FUTURE DIRECTIONS

The following general guidelines provide a general framework for using post-quantum cryptography. Each organization's requirements and circumstances may differ, so it's important to adapt these guidelines to our specific needs and consult with experts when necessary.

1) **Staying informed:** We should Keep ourselves up to date with the latest developments in post-quantum cryptography by following reputable sources such as NIST and academic research papers. We should also stay informed about new algorithms, their security properties, and any updates or advancements in the field.
2) **Understanding threat model:** We should assess our organisation's specific risks and threats while considering factors such as sensitive data protection, potential adversaries, and the anticipated lifespan of our cryptographic systems. This understanding will help us determine the appropriate level of security and select the most suitable post-quantum cryptographic algorithms.
3) **Evaluating algorithms:** We should thoroughly evaluate different post-quantum cryptographic algorithms while considering security, performance, key size, and implementation complexity factors. We should also assess the algorithm's resistance to classical and quantum attacks and ensure it aligns with our organization's security requirements and computational resources.
4) **Standardization efforts:** We should consider algorithms that are part of standardization processes, such as those being evaluated in competitions like the NIST PQC competition. Standardization ensures interoperability and facilitates the adoption of post-quantum cryptography across different systems and platforms. However, we should also be open to considering new algorithms that show promise but may not have reached the standardization stage yet.
5) **Assessing implementation feasibility:** We should evaluate the practical feasibility of implementing the chosen post-quantum cryptographic algorithms within our systems and infrastructure. We should consider factors such as available libraries, support for different platforms, and the expertise required for integration, taking into account performance considerations, such as computational overhead and memory requirements, to ensure the efficient operation of our systems.
6) **Planning for a smooth transition:** We should develop a well-defined transition plan from classical to post-quantum cryptographic systems. This includes identifying critical systems and data that require immediate protection, establishing a timeline for implementation, and ensuring compatibility with existing systems and protocols. Consider the potential impact on applications, user experience, and third-party integrations during the transition process.
7) **Implementing a hybrid approach:** We should consider adopting a hybrid cryptographic approach that combines classical and post-quantum cryptographic algorithms. This approach is compatible with existing systems while offering protection against quantum attacks. Hybrid schemes can secure key exchange, digital signatures, and other cryptographic operations until complete migration to post-quantum algorithms is feasible.
8) **Engaging with the community:** We should participate in the post-quantum cryptography community to share knowledge, experiences, and best practices and should engage with researchers, industry experts, and standards bodies to stay updated on the latest advancements and receive guidance on implementation and deployment strategies.
9) **Regular reassess and update:** Post-quantum cryptography is a rapidly evolving field, and discoveries, advancements, and attacks may emerge over time. Considering the latest research and developments, we should regularly reassess cryptographic systems and algorithms. We must stay proactive in updating our systems to maintain security against emerging threats.

X. CONCLUSION

Post-Quantum Cryptography (PQC) is a vital field of research and development aimed at addressing the potential threat quantum computers pose to the security of current cryptographic algorithms. This paper has explored the motivation behind PQC, presented experimental research conducted in the field, and provided guidelines for its adoption.

The motivation for PQC stems from the realization that quantum computers have the potential to break widely used classical cryptographic algorithms, jeopardizing the security of sensitive information. The vulnerability of algorithms like RSA and ECC to quantum attacks necessitates searching for and adopting quantum-resistant alternatives.

The widespread adoption and implementation of post-quantum cryptography (PQC) has yet to progress. The PQC-PKI adoption could be accelerated once PQC algorithms are standardized and validated by the community. The progress made in implementing PQC-based PKI worldwide is expected

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to be gradual because of the vast digital infrastructures and communications systems that rely on traditional cryptographic algorithms. Adopting PQC PKI will require significant coordination and planning across government, industries, and organizations.

In conclusion, post-quantum cryptography is an essential response to the growing threat of quantum computing. The research, experiments, and guidelines discussed in this paper contribute to understanding PQC and its practical implementation. By proactively preparing for the post-quantum era and adopting quantum-resistant cryptographic algorithms, organizations can ensure the long-term security and integrity of their sensitive data and communications in the face of advancing quantum technologies.

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